



SplicR for the Varmus lab

A549 LKB1 screens, revisited

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The paper's biological question

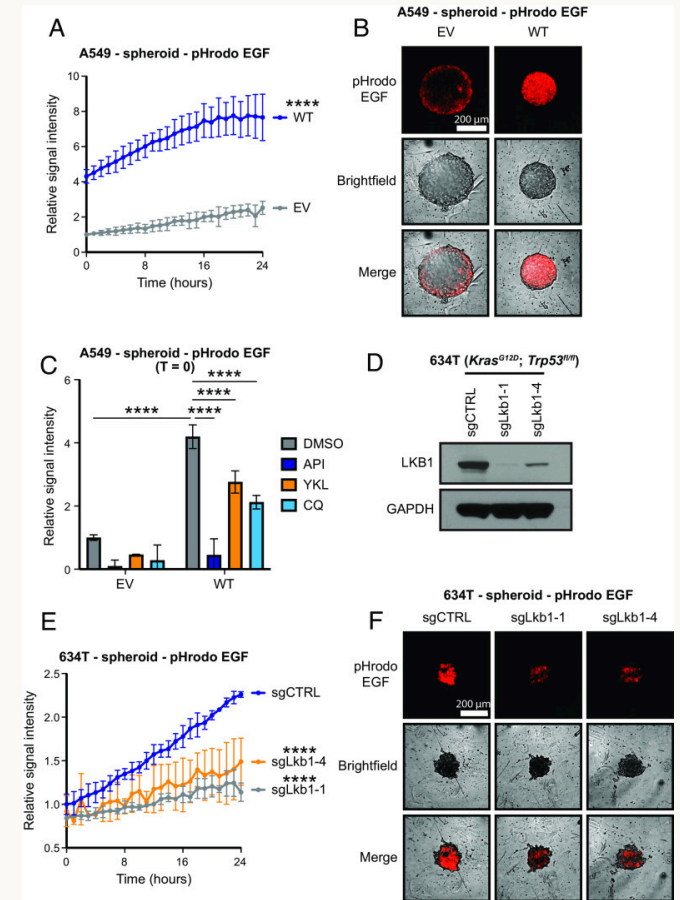
How does LKB1 restrain lung cancer cell growth in spheroids?

The screen points to the PIKFYVE complex.

Follow-up experiments connect it to how cells take in growth signals.

SplicR examines the CRISPR evidence behind that story.

Ferrarone et al., PNAS 2024, Fig. 4. Published experiment, not a SplicR output.



The screen design

A549 cells, TKOv3 knockout library, day 21

Culture	LKB1-null (EV)	LKB1-restored (WT)
2D monolayer	a and b	a and b
3D spheroid	a and b	a and b

One shared plasmid library reference

Eight endpoint samples. The plasmid reference is not a biological replicate.

The deposited files pass identity checks

Both original downloads match Harvard Dataverse V1

70,116

observed guides in each table

18,049

genes after guide-ID annotation

104 date-like gene labels corrected per file

Original counts remain unchanged. The analysis uses TKOv3 annotations keyed by guide ID.

Upload the original count tables

Two files.

No FASTQ download is needed for this count-table demo.

	2d_crispr_screen_read_counts.txt	COUNTS	3.9 MB	sha256 b5e01844	✓ Uploaded	×
▶ Inspect file						
	spheroid_crispr_screen_read_counts.txt	COUNTS	3.9 MB	sha256 efa79d1c	✓ Uploaded	×
▶ Inspect file						

SplicR identifies the guide library

All observed guide IDs
match TKOv3.

The mapping stays
visible for review.

Guide library

TKOv3, 71,090 guides

▼

Upload library

70,116 of 70,116 checked guide IDs match.

Recheck library

Library name for this experiment

TKOv3

Six comparisons, ready to review

EV versus plasmid.
WT versus plasmid.
WT versus EV.

Separately for each
culture.

☒ 2D: EV vs plasmid	2 vs 1 samples, RRA + MLE
☒ 2D: WT vs plasmid	2 vs 1 samples, RRA + MLE
☒ 2D: WT vs EV	2 vs 2 samples, RRA + MLE
☒ Spheroid: EV vs plasmid	2 vs 1 samples, RRA + MLE
☒ Spheroid: WT vs plasmid	2 vs 1 samples, RRA + MLE
☒ Spheroid: WT vs EV	2 vs 2 samples, RRA + MLE

Researchers control the analysis

Median normalization.
FDR threshold 0.05.
RRA with optional MLE.

The reviewed plan
travels with the run.

Pipeline settings

Readout

Pooled guide abundance

Model type

cancer cell line

FDR threshold

0.05

Normalization

Median ratio

▼ Advanced settings

☒ Add MAGeCK MLE

☐ Add BAGEL2 to fitness comparisons

☐ Paired DrugZ, matching replicate numbers

DrugZ pseudocount

5

DrugZ smoothing half-window

500

Chronos requires pDNA batches and collection times. Copy-number correction requires matched profiles. These methods are unavailable in this intake.

Live progress follows recorded stages

All six full RRA + MLE analyses are complete.

C

Analyzing: Ferrarone 2024: A549 LKB1 growth screens, 2D: EV vs plasmid

4 of 9 stages complete , Elapsed: 1h 35m 53s

Analysis worker active.

Hits

Running MAGeCK MLE...

Cancel analysis

✓

Ingest

✓

Detect

✓

Count

✓

QC

5

Hits

6

Artifacts

7

Atlas

8

Score

9

Report

Captured during MLE fitting. All six full RRA + MLE analyses are now complete.

09

A completed report exposes uncertainty

Spheroid WT versus plasmid, completed RRA + MLE run

Screen summary run 01a11702			
Warning Quality control Passed with something worth checking.	178 Candidates At or below FDR 0.05. A record with no recorded FDR is not counted.	172 of 178 With artifact flags A flag is a reason to check a candidate, not a verdict on it.	None Validation No gene from this screen has been taken to the bench yet.
Screen health QC warn			
The assay worked, but 1 check failed.			
LOOK AT THIS FIRST Replicates disagree Review wt_3d_d21_a and wt_3d_d21_b before interpreting the arm they belong to.			
> What was checked			1 of 2 need attention
> Every sample's recorded QC			3 samples

178 candidates. 172 flagged for review, not automatic rejection.

176 depleted. 154 annotated pan-essential in DepMap, consistent with common survival dependencies.

Results across the six comparisons

RRA candidates at the declared two-direction FDR threshold of 0.05



Each report covers 18,049 genes. These RRA counts do not reproduce the paper's MLE figure.

The paper's genes, across both methods

Spheroid WT versus plasmid, complete recorded output

Gene	RRA log ₂ change	MLE beta	MLE FDR
FIG4	+3.14	+1.83	0.576
VAC14	+4.96	+3.21	0.192
PIKFYVE	+0.41	+0.44	1.000

Positive effects and significance are different questions

None passes MLE FDR 0.05 here. The paper's beta-score nomination uses different criteria.

The export keeps the underlying evidence

CSV for analysis. JSON and receipts for reproducibility.

Ordered by recorded FDR. Engine 652db059378a+ce7ef8095972.			
1-100 of 18,049 Previous Page 1 of 181 Next Export CSV			
Analysis provenance run 01a11702			
Run status complete	Engine version 652db059378a+ce7ef8095972	Container digest Not recorded	Run ID 01a11702-77d1-76fb-88ac-039c02f5d0eb
Run saved analysis again			
ingest done			
1 file(s), 0.00 GB, wt_3d_d21_a+wt_3d_d21_b vs tkov3_plasmid			
detect done			
TKOv3 (supplied, not detected)			

18,049 gene rows
per comparison

Inputs, settings and
quality warnings remain
attached to the run.

The demo includes the actual downloaded CSV and complete receipts for all six RRA + MLE runs.

Measured runtime and reproducibility

Completed RRA batch, original deposited count tables

6 min 2 sec

First worker start to last worker finish across six comparisons

97–126 sec per worker

Local and cloud primary statistics agree exactly

RRA timing shown. Initial full MLE cohort: 5 h 5 min, including retries. Upload time excluded.

SplicR alongside Broad Apron

Apron already offers a strong guided analysis workflow

This study	Broad Apron guide	SplicR demonstration
Starting files	PoolQ counts and annotations Other inputs need formatting	Original deposited count tables
Design setup	User defines references and data configurations	Six editable suggestions from verified source identity
Gene statistics	Control-set z-scores and empirical null	MAGeCK RRA Optional separate MLE
Annotations and evidence	Guide describes QC and analysis exports	Guide-ID annotation repair Flags beside recorded results

Documented workflow comparison. No head-to-head accuracy or runtime benchmark.

Other tools and SplicR's role

Established methods remain the statistical foundation

Tool	Established strength	SplicR's role in this demo
MAGeCK	RRA and MLE Explicit experimental designs	Guided setup, hosted compute and reviewable reports
PinAPL-Py	Web analysis, custom libraries Automated read extraction	Shared workspace context and source-linked run receipts
Broad Apron	Guided QC and control-set gene aggregation	Verified study intake and guide-ID annotation repair

SplicR has no demonstrated universal accuracy advantage over these methods.

A reusable workflow for other labs

Researchers can
change sample roles,
replicates and factors.

Their experiment
determines the
configuration.

SAMPLE	ROLE	REPLICATE	MODEL	CONDITION	TIMEPOINT
tkov3_plasmid	Plasmid ▼	1	A549	TKOv3 plasmid	plasmid
ev_2d_d21_a	Control ▼	1	A549	LKB1-null (EV)	21
ev_2d_d21_b	Control ▼	2	A549	LKB1-null (EV)	21
wt_2d_d21_a	Endpoint / treatment ▼	1	A549	LKB1 restored (W)	21
wt_2d_d21_b	Endpoint / treatment ▼	2	A549	LKB1 restored (W)	21

A pilot with the Varmus lab

A reproducible starting point
for the next screen

Agree on the reference analysis.
Review differences together.
Test candidate genes in the lab.

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Methods and scientific limits

Appendix for scientific discussion

Completed evidence

Original counts and verified guide annotations

Six RRA + MLE reports with separate statistics

Independent RRA numerical reproduction

Open scientific work

Reconcile MLE with the authors' settings

Fit a formal culture interaction if required

Validate candidates experimentally

Validation probabilities remain unavailable without a fitted outcomes cohort.

Sources and demo files

Appendix for follow-up

Paper

10.1073/pnas.2403685121

SplicR-Ferrarone-demo.zip

Two original count tables

CRISPR data

10.7910/DVN/8DEPIT

SplicR-Ferrarone-full-results.zip

Six complete RRA + MLE result packages

[Open full spheroid WT report](#)

Report requires sign-in to the retained demo workspace. All six full RRA + MLE packages are included.